EE2080 : Lecture 13 : 26 August 2025

PLAN:
- Gaussian Processes & Brownian Motion
- Stopping times w.r.t Filtrations
- Ergodicity of a Random Process
- WSS processes through an LTI System

REF:
1) R. Gallager, "Stochastic Processes: Theory for Applications
2) H. Stark, J. W. Woods, "Probability, Statistics and Random Processes for Engineers".
3) B. Hajek, "Random Processes for Engineers".

Gaussian Random Variables

- $X \sim N(0,1)$ if $f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-x^2}{2}\right)$ $\quad x \in (-\infty, \infty)$. Standard Normal

mean = 0, variance = 1

- $Y = \sigma X + \mu$, $Y \sim N(\mu, \sigma^2)$ $\quad f_Y(y) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right)$, $y \in (-\infty, \infty)$

Y - Gaussian RV with mean μ & variance σ^2.

- MGF of Y: $M_Y(s) = \mathbb{E}\left[\exp^{sY}\right] = \exp\left(s\mu + s^2\sigma^2/2\right)$

Gaussian Random Vectors: $\underline{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} \sim N(\underline{\mu_X}, K) \Rightarrow X_1, X_2, \ldots, X_n$ are jointly Gaussian R. Variables.

$$f_{\underline{X}}(\underline{x}) = \frac{1}{\sqrt{(2\pi)^N \det(K)}} \exp\left[\frac{-1}{2}(\underline{x} - \underline{\mu_x})^T K^{-1} (\underline{x} - \underline{\mu_x})\right] \quad \forall \underline{x} \in \mathbb{R}^n$$

K = Covariance matrix $\mathbb{E}\left[(\underline{X} - \underline{\mu_X})(\underline{X} - \underline{\mu_X})^T\right]$; $K_{i,j} = \mathbb{E}\left[(X_i - \mu_i)(X_j - \mu_j)\right]$

Gaussian Process: A stochastic process $\{X_t, t \in T\}$ is said to be a Gaussian process if for all $k \in \mathbb{N}$ and $t_1, t_2, \ldots, t_k \in T$, the set of random variables $X_{t_1}, X_{t_2}, \ldots, X_{t_k}$ is jointly Gaussian.

The covariance function: $C_X(s,t) = \mathbb{E}\left[(X_s - \mu_X(s))(X_t - \mu_X(t))\right]$, $\forall s,t \in T$

For a Gaussian process $\{X_t, t\in T\}$, the covariance function $C_X(s,t)$ and the mean $\mu_X(t)$ for all $s,t \in T$ completely specify the joint-pdf for all $(X_{t_1}, X_{t_2}, \ldots, X_{t_k})$, $\forall k \in \mathbb{N}$, $t_1, t_2, \ldots, t_k \in T$.

Example 1: Discrete-time IID Gaussian process

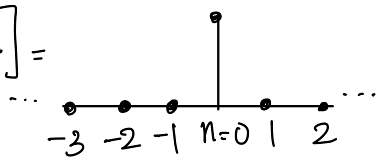
$\{W_n, n \in \mathbb{Z}\}$, $W_n \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$

$$C_W(k,l) = \sigma^2 \delta[l-k]$$

$$f_{W_{n_1} \ldots W_{n_k}}(w_1, \ldots, w_k) = \frac{1}{(2\pi\sigma^2)^{k/2}} \exp\left(-\sum_{i=1}^{k} \frac{w_i^2}{2\sigma^2}\right)$$

Discrete-time delta-fn.

$\delta[n] =$



Example 2: Discrete-time Gaussian sum-process.

$$S_n = \sum_{i=1}^{n} W_i \quad \longrightarrow \text{linear combination of iid Gaussian RVs.}$$

$\{S_n, n \geq 1\}$ - zero-mean Gaussian process.

For $k \leq l$, $C_S(k,l) = \mathbb{E}\left[\sum_{i=1}^{k} W_i \sum_{j=1}^{l} W_j\right] = \sum_{i=1}^{k} \mathbb{E}[W_i^2] = k\sigma^2$

For any k,l $C_S(k,l) = \min(n,k)\,\sigma^2$.

Theorem: A Gaussian process $\{X_t, t\in T\}$ is stationary if and only if $\mu_X(t) = \mu_X(0)$ is a constant and $C_X(s,t) = C_X(0, t-s)$ $\forall t,s \in T$

$\Rightarrow$ A W.S.S Gaussian process is stationary!

The Wiener process / Brownian Motion

Consider a stochastic process $\{X_t, t\in T\}$ where $T = \mathbb{Z}^+$ or $T = \mathbb{R}$.

i) $\{X_t, t\in T\}$ has stationary increments if for any $t_1 < t_2$, $X(t_2) - X(t_1) \overset{d}{=} X(t_2 - t_1) - X(0)$

ii) $\{X_t, t\in T\}$ has independent increments if for any $t_1 < t_2 < t_3 < \dots < t_k$, the RVs $X_{t_2} - X_{t_1}, X_{t_3} - X_{t_2}, \dots \ X_{t_k} - X_{t_{k-1}}$ are statistically independent

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Defn: A Wiener Process / Brownian motion is a zero-mean Gaussian process $\{X_t, t \geq 0\}$ which has the following properties

i) Stationary and independent increments

ii) $X_0 = 0$ with probability 1.

iii) has continuous sample functions with prob. 1, i.e X_t is almost surely continuous in t.

Stopping Time

Fix a measurable space $(\Omega, \mathcal{F})$

Let $X : \Omega \to \mathbb{R}$ be a random variable w.r.t $\mathcal{F}$

Defn: The σ-algebra generated by X, denoted $\sigma(X)$ is defined as

$$\sigma(X) = \{A \in \mathcal{F} : A = X^{-1}(B) \text{ for some } B \in \mathcal{B}(\mathbb{R})\}$$

Example: $\Omega = \{1, 2, \dots, 6\}$ $\mathcal{F} = 2^{\Omega}$

$$X(\omega) = \begin{cases} 0, & \omega = 1,2 \\ 4, & \omega = 5,6 \\ 6, & \omega = 3,4 \end{cases}$$

$$X^{-1}([-\infty, x]) = \begin{cases} \phi, & x < 0 \\ \{1,2\}, & 0 \leq x < 4 \\ \{1,2,5,6\}, & 4 \leq x < 6 \\ \Omega, & x \geq 6 \end{cases}$$

$$\sigma(X) = \{\phi, \{1,2\}, \{3,4\}, \{5,6\}, \{1,2,3,4\}, \{1,2,5,6\}, \{3,4,5,6\}, \Omega\}$$

Remark: $\sigma(X)$ is the smallest σ-algebra w.r.t which X is a R.V.

σ-Algebra generated by a random vector: Let $(X_1, \ldots, X_n) : \Omega \to \mathbb{R}^n$ be a random vector w.r.t $\mathcal{F}$.

$$\sigma(X_1, \ldots, X_n) = \left\{ A \in \mathcal{F} : A = (X_1, \ldots, X_n)^{-1} B \text{ for some } B \in \mathcal{B}(\mathbb{R}^n) \right\}$$

Filtrations: Fix a prob. space $(\Omega, \mathcal{F}, \mathbb{P})$

Let $\mathcal{T}$ be an ordered index set

Consider a collection of σ-algebras $\mathcal{G}_\cdot = \{\mathcal{G}_t : t \in \mathcal{T}\}$ s.t $\mathcal{G}_t \subseteq \mathcal{F} \; \forall t$.

The above collection is called a filtration if $\mathcal{G}_s \subseteq \mathcal{G}_t \quad \forall s \leq t$

Example: Let $\{X_t : t \in \mathcal{T}\}$ be a random process defined w.r.t $\mathcal{F}$. Then $\mathcal{G}_t = \sigma(X_s : s \leq t)$ is called the natural filtration associated with the process $\{X_t : t \in \mathcal{T}\}$.

Stopping Time:

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $\mathcal{T}$ be an ordered index set.

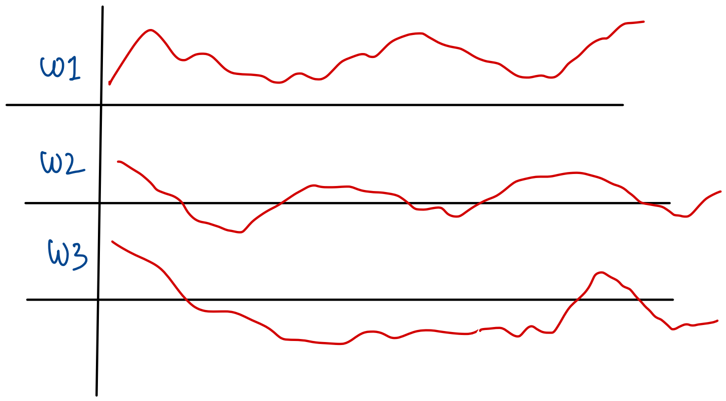
Fix a filtration $\mathcal{G}_\cdot = \{\mathcal{G}_t : t \in \mathcal{T}\}$.

Defn: A random variable $T : \Omega \to \mathbb{R} \cup \{\pm\infty\}$ is called a stopping time w.r.t the filtration $\mathcal{G}_\cdot$ if

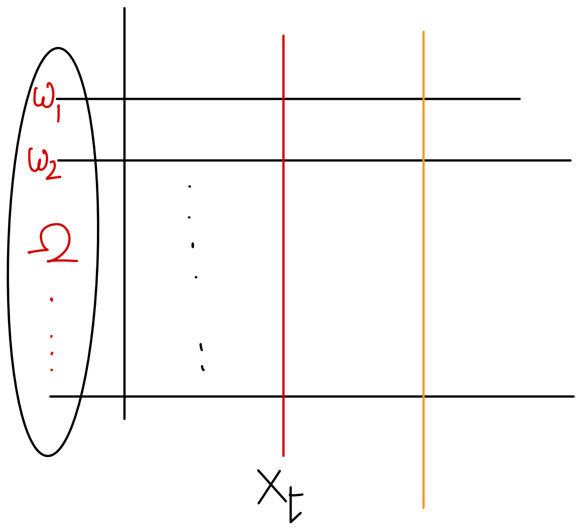
- $\mathbb{P}(T < +\infty) = 1$
- For each $t \in \mathcal{T}$, $\{T \leq t\} \in \mathcal{G}_t$.

That is, the answer to the question "is $T \leq t$?" can be answered by looking at the process upto (and including) time t.

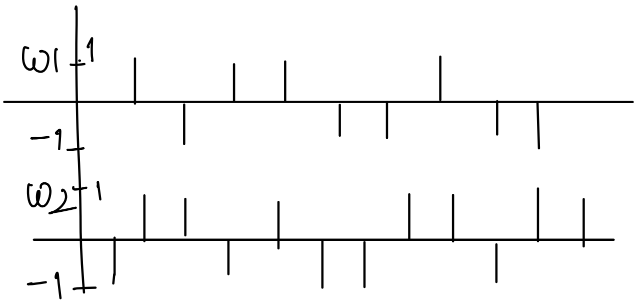
Ergodicity



$\{X_t, t \geq 0\}$ is a continuous valued continuous parameter process. Assume that it is stationary.
From a single path, is it possible to deduce the statistical properties of the process?



What about this process $\{X_n, n \geq 0\}$



Ensemble average: $\mu_X(t) = \mathbb{E}[X_t] = \int x f_X(x)\,dx$ or $\sum x p_X(x)$.

Time Average (temporal) $\hat{\mu}_X(T) = \frac{1}{T}\int_{-T/2}^{T/2} X_\omega(t)\,dt$ $\quad \hat{\mu}_X = \lim_{T\to\infty} \hat{\mu}_X(T)$

Time averaged autocorrelation $\hat{R}_{X_T}(\Delta) = \frac{1}{T}\int_{-T/2}^{T/2} X_\omega(t)\, X_\omega(t+\Delta)\,dt$. $\quad \hat{R}_X(\Delta) = \lim_{T\to\infty} \hat{R}_{X_T}(\Delta)$

Ensemble averaged autocorrelation: $R_X(t,\Delta) = \mathbb{E}[X_t X_{t+\Delta}]$.

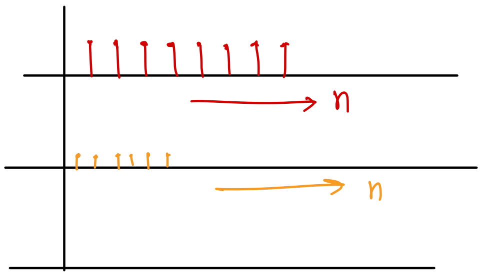
A stationary random process is said to be ergodic in mean if its time average converges to the ensemble average in the mean square sense: $\hat{\mu}_{X_T} \xrightarrow{m.s} \mathbb{E}[X]$ i.e $\hat{\mu}_X = \mathbb{E}[X]$

A stationary random process is said to be ergodic in autocorrelation if its time-averaged autocorrelation converges to the ensemble-averaged autocorrelation in the mean square sense.
$\hat{R}_{X_T}(\Delta) \xrightarrow{m.s} R_X(\Delta)$ i.e., $\hat{R}_X(\Delta) = R_X(\Delta)$.

Stationary but not ergodic: Example $(\Omega, \mathcal{F}, \mathbb{P})$

$\{X_n, n \in \mathbb{N}\}$ s.t For $\omega \in \Omega$, $X_n(\omega) = \omega \quad \forall n \in \mathbb{N}$

Is this process stationary? Is it ergodic?

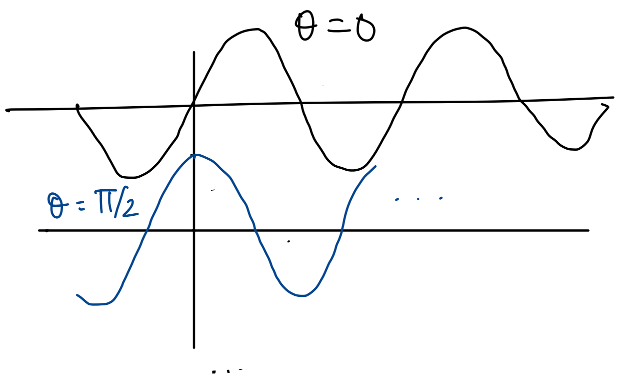


Example of an ergodic process

$X_t = A\sin(\omega_0 t + \Theta) \qquad \Theta \sim \text{unif}(0, 2\pi) \qquad A, \omega_0$ fixed positive constants

$$\hat{\mu}_{X_\theta}(T) = \frac{1}{T}\int_{-T/2}^{T/2} A\sin(\omega_0 t + \theta) \quad = \underline{\underline{0}}$$

for some $\theta \in [0, 2\pi]$



$$\mathbb{E}[X_t] = \int_0^{2\pi} A\sin(\omega_0 t + \theta)\frac{1}{2\pi}\,d\theta = 0$$

Properties of Autocorrelation:

Consider a W.S.S Process $\{X_t, t \in T\}$

1) $|R_{X,X}(\tau)| \leq R_{X,X}(0)$

$$\left|\mathbb{E}[X_t X_{t+\tau}]\right| \leq \sqrt{\mathbb{E}[X_t^2]\,\mathbb{E}[X_{t+\tau}^2]} \qquad \text{Cauchy Schwarz}$$

$$\sqrt{R_X(0)\,R_X(0)}$$

$$|R_X(\tau)| \leq R_X(0)$$

2) Assuming X_t is real, $R_X(\tau)$ is an even function of τ, i.e., $R_X(\tau) = R_X(-\tau)$

$$R_X(\tau) = \mathbb{E}[X_t X_{t-\tau}]$$

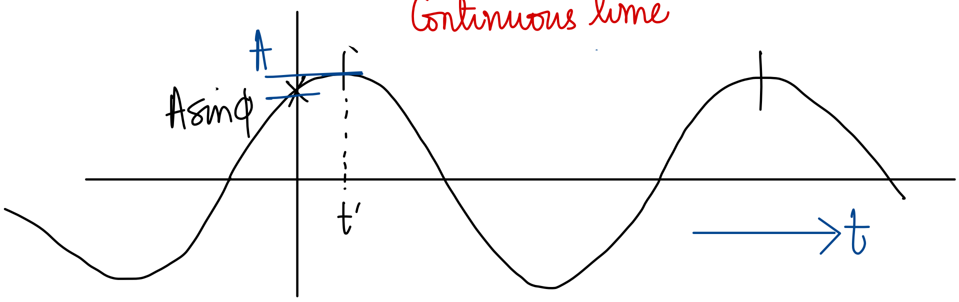
$$R_X(-\tau) = \mathbb{E}[X_{t-\tau} X_t]$$

WSS Processes through LTI S/ms

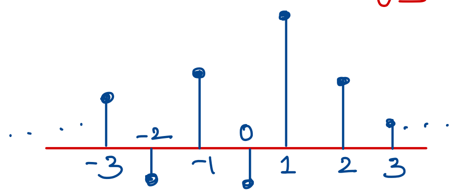
Linear Time-Invariant Systems :

Signals : Example : $x(t) = A\sin(\omega_0 t + \phi)$

Continuous time

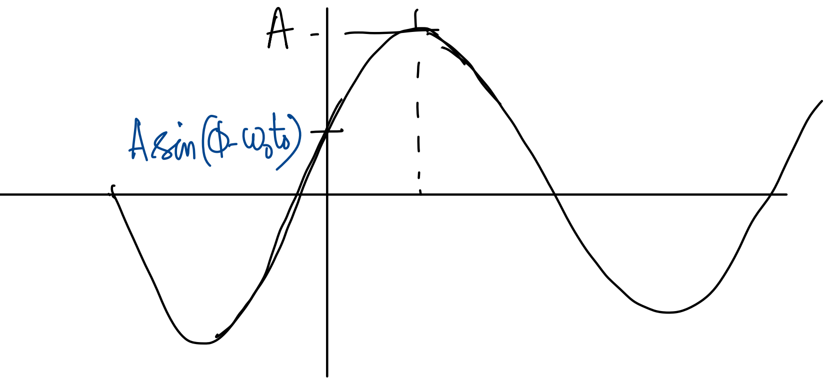


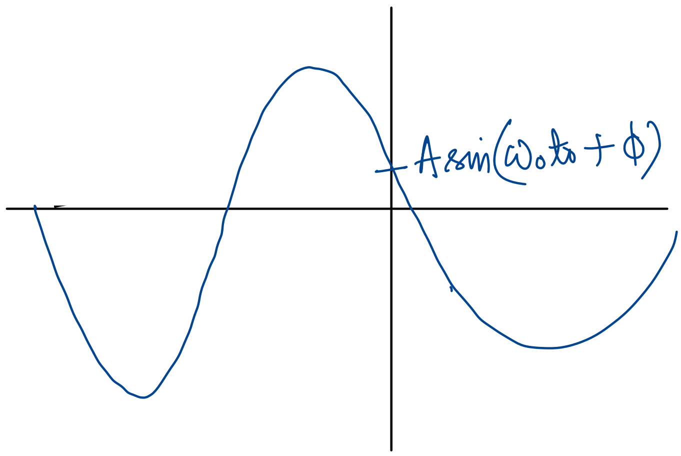
Discrete time $x[n]$



Time-Shift : $x(t-t_0)$: $A\sin(\omega_0(t-t_0)+\phi)$

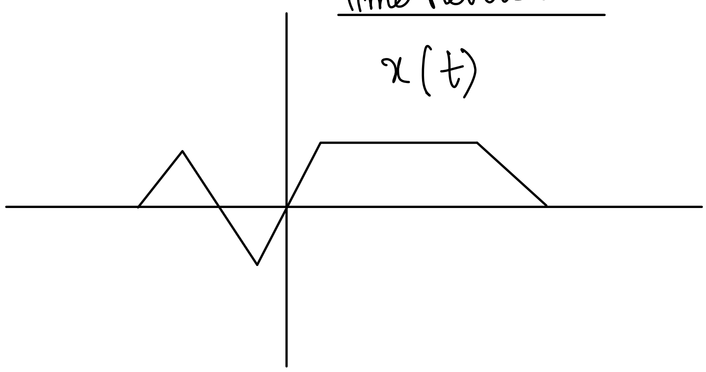
$x(t+t_0) = A\sin(\omega_0 t + \omega_0 t_0 + \phi)$



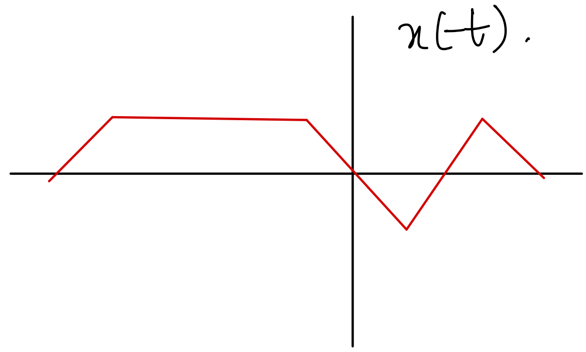


Time Reversal

$x(t)$



$x(-t)$.



Power and Energy of Signals

Total energy of a signal $x(t)$ in the time interval $t_1 \le t \le t_2 = \int_{t_1}^{t_2} |x(t)|^2 dt$.

Time-average power : $\frac{1}{t_2 - t_1} \int_{-\infty}^{\infty} |x(t)|^2 dt$.

For $x[n] = \underbrace{\sum_{n=n_1}^{n_2} |x[n]|^2}_{\text{energy}}$. $\quad \underbrace{\frac{1}{n_2 - n_1 + 1} \sum_{n=n_1}^{n_2} |x[n]|^2}_{\text{Power}}$

If the signal extends over an infinite time interval

$$E = \lim_{T\to\infty} \int_{-T}^{T} |x(t)|^2 dt \qquad E = \lim_{N\to\infty} \sum_{n=-N}^{N} |x[n]|^2 .$$ Energy

$$P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 dt \qquad P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2 .$$ Power

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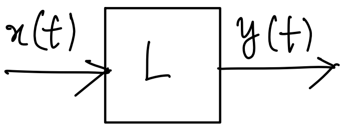
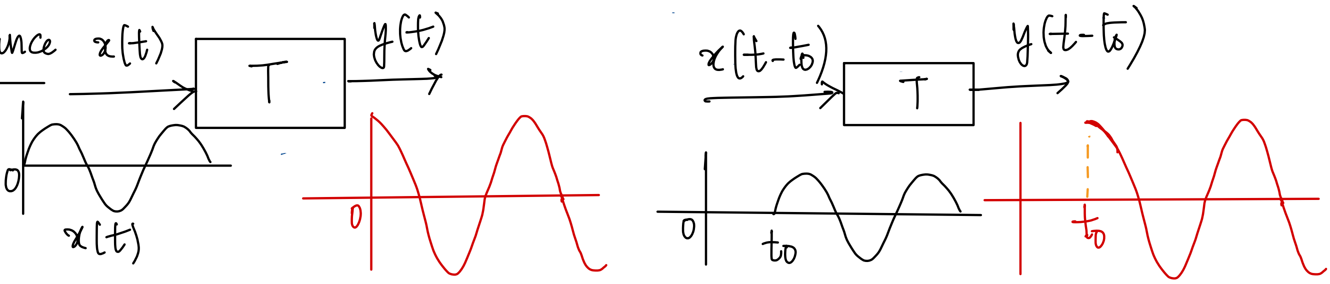
Linear Systems

$$y(t) = L[x(t)]$$



$$L[a_1 x_1(t) + a_2 x_2(t)] = a_1 L[x_1(t)] + a_2 L[x_2(t)] .$$

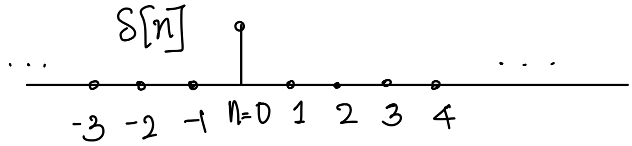
Time Invariance



If we apply an input to the system now or t_0 seconds from now, the output will be identical except for a delay of t_0 seconds; i.e., the output shape does not depend on the particular time at which the input is applied.

LTI systems are characterised by their impulse response.

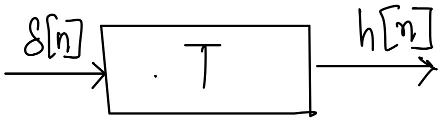
Discrete time impulse signal



$$\delta[n] = \begin{cases} 1, & n=0 \\ 0, & n\neq 0 \end{cases}$$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k] \qquad \delta[n-k] = \begin{cases} 1, & n=k \\ 0, & n\neq k \end{cases}$$

T: LTI system: Suppose

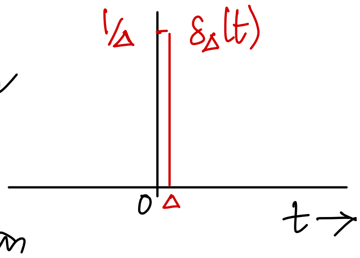


Impulse Response.

Then $$T[x[n]] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$ Discrete-time LTI System

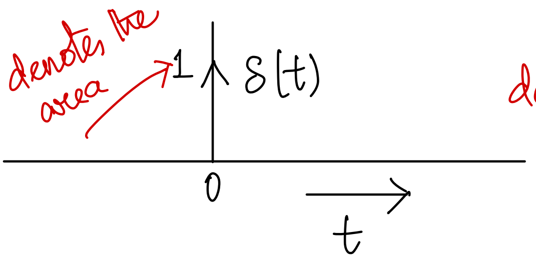
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$ Convolution Sum.

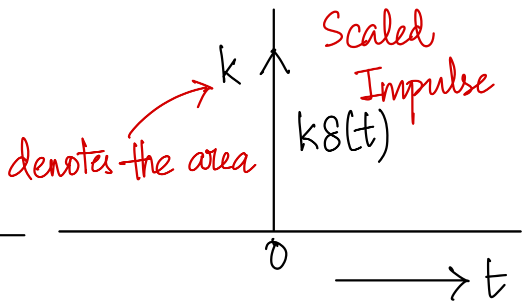
Continuous-time impulse signal

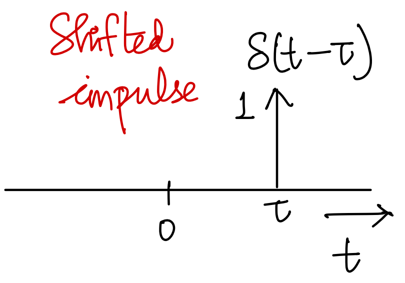


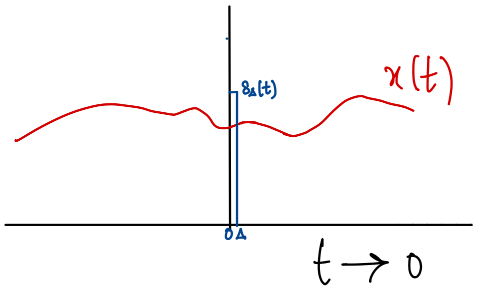
Consider a pulse $\delta_\Delta(t)$ of duration Δ and unit area as shown in figure. As Δ decreases, $\delta_\Delta(t)$ becomes narrower and higher maintaining its unit area. Its limiting form $\delta(t)$ defined as

$\delta(t) = \lim_{\Delta \to 0} \delta_\Delta(t)$ is the continuous-time impulse signal









Consider $x_1(t) = x(t)\,\delta_\Delta(t)$.

It is zero outside $0 \le t \le \Delta$.

For a sufficiently small Δ s.t $x(t)$ is constant in $[0, \Delta]$,

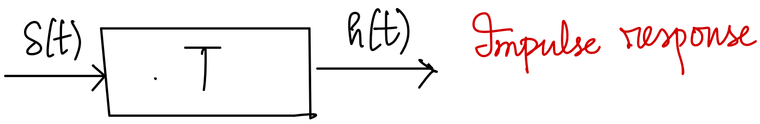
$$x(t)\,\delta_\Delta(t) = x(0)\,\delta_\Delta(t).$$

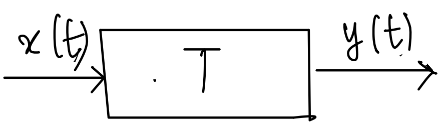
$\because\ \delta(t) = \lim_{\Delta \to \infty} \delta_\Delta(t)$, we have $x(t)\delta(t) = x(0)\,\delta(t)$.

Similarly, $x(t)\,\delta(t-\tau) = x(\tau)\,\delta(t-\tau)$

So, similar to a discrete-time LTI S/m, a continuous-time LTI system can be completely characterised by its response to the CT-impulse signal, called the impulse response denoted $h(t)$.

Consider a CT LTI S/m T:





$$y(t) = x(t) * h(t)$$
$$= \int x(\tau)\, h(t-\tau)\, d\tau$$

Convolution integral.

Fourier Transform of a Signal

Frequency domain representation of a signal.

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

Fourier transform of $x(t)$
(Analysis Equation)

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega.$$

(Synthesis equation)

$x(t)$ as a linear combination of complex exponentials at different frequencies

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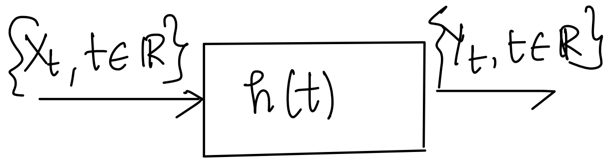
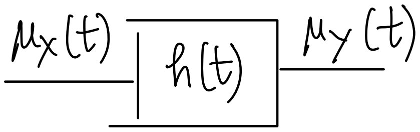
Random Processes through LTI Systems



$$Y_t = \int_{-\infty}^{\infty} h(t-\tau)\, X_\tau\, d\tau$$

Mean process : $$\mu_Y(t) = \mathbb{E}\left[\int_{-\infty}^{\infty} h(t-\tau) X_\tau\, d\tau\right] = \int h(t-\tau)\, \underbrace{\mathbb{E}[X_\tau]}_{\mu_X(\tau)}\, d\tau$$



Suppose that $\{X_t : t \in \mathbb{R}\}$ is WSS.

$$\mu_Y(t) = \int_{-\infty}^{\infty} h(t-\tau)\, \mu_X\, d\tau = \mu_X \underbrace{\int_{-\infty}^{\infty} h(t-\tau)\, d\tau}_{\underbrace{\int_{-\infty}^{\infty} h(t)\, dt}_{H(0)}}.$$

Fourier transform of the impulse response

$$H(\omega) = \int_{-\infty}^{\infty} h(t) e^{-j\omega t}\, dt$$

$$H(0) = \int_{-\infty}^{\infty} h(t)\, dt.$$

$$\boxed{\mu_Y = \mu_X \cdot H(0)}$$

Jointly-WSS Random Processes

$\{X_t, t \in T\}$ and $\{Y_t, t \in T\}$ are said to be jointly WSS if i) each process is WSS

ii) $R_{X,Y}(t,s) = \mathbb{E}[X_t, Y_s]$ is a function of $(t-s)$ alone.

Cross-Correlation between o/p and i/p

$$R_{Y,X}(t,s) = \mathbb{E}[Y_t X_s] = \mathbb{E}\left[\int_{-\infty}^{\infty} h(t-\tau) X_\tau \, d\tau \cdot X_s\right]$$

$$= \int_{-\infty}^{\infty} h(t-\tau) \underbrace{\mathbb{E}[X_\tau X_s]}_{R_X(\tau-s)} d\tau$$

$$\boxed{R_{Y,X}(t,s) = (h * R_X)(t-s)}$$

$$R_{Y,Y}(t,s) = \mathbb{E}\left[Y_t \int_{-\infty}^{\infty} h(s-u) X_u \, du\right]$$

$$= \mathbb{E}\left[\int_{-\infty}^{\infty} h(t-\tau) X_\tau \, d\tau \int_{-\infty}^{\infty} h(s-u) X_u \, du\right]$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(t-\tau)\, h(s-u)\, R_X(\tau-u)\, d\tau\, du$$

$$= \int_{-\infty}^{\infty} h(s-u) \underbrace{\left(\int_{-\infty}^{\infty} h(t-\tau) R_X(\tau-u)\, d\tau\right)}_{R_{YX}(t-u) = R_{YX}(u-t) \text{ even fn}} du$$

$$= \int_{-\infty}^{\infty} h(s-u) R_{YX}(u-t)\, du$$

$$R_{YY}(t-s) = (h * R_{Y,X})(s-t)$$

$\therefore$ If a W.S.S process $\{X_t, t \in T\}$ is passed through an LTI s/m, the output process $\{Y_t, t \in T\}$ is jointly-W.S.S with the input process

Consider a W.S.S process $\{X_t, t \in T\}$

The power spectral density $S_{X,X}(\omega)$ of the process $\{X_t, t \in T\}$ is defined as the Fourier transform (if it exists) of its autocorrelation function $R_X(\tau)$

i.e., $$S_{X,X}(\omega) = \int_{-\infty}^{\infty} R_X(\tau)\, e^{-j\omega\tau}\, d\tau .$$

We can show that for an LTI s/m $\{X_t, t \in T\} \longrightarrow \boxed{h(t)} \longrightarrow \{Y_t, t \in T\}$

$$S_{Y,Y}(\omega) = |H(\omega)|^2 \cdot S_{X,X}(\omega)$$